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Brief paper

Zonotopic state estimation and fault detection for discrete-time linear systems under DoS attacks: A switching controller design mechanism ☆

Zhihua Guo, Xinjun Wang *, Mingliang Tian, Jianing Hu

School of Computer Science and Artificial Intelligence, Shandong Normal University, Jinan 250014, China

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ABSTRACT

This paper investigates the issues of zonotopic-based set-membership state estimation and fault detection for discrete-time linear systems with unknown but bounded (UBB) disturbance and noise under nonperiodic denial-of-service (DoS) attacks. Firstly, a novel zonotopic state observer is designed to obtain point-valued estimations for the considered system. To improve estimation accuracy, a decoupling method is simultaneously proposed to separate the disturbance from the error dynamics. Secondly, the discrete-time linear systems subjected to intermittent nonperiodic DoS attacks are restructured into a class of augmented switched systems, which comprises both a stable subsystem and an unstable subsystem. To guarantee the stability of the augmented system, a switching law is designed, which solves the instability problem of system under long-term or high-frequency DoS attacks. Based on Lyapunov stability theory, the exponential stability analysis and l_2-gain performance analysis are presented for the augmented switched system. Moreover, the reachable set of the system state under nonperiodic DoS attacks and fault-free scenarios is obtained through reachability analysis. In addition, a more reliable set-membership fault detection strategy with the obtained reachable set and the residual signals is developed. Finally, some simulation results are provided to show the advantages of the theoretic results.

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1. Introduction

The rapid development of communication, control theory and artificial intelligence technology boosts the emergence and application of discrete-time linear cyber–physical systems (CPSs). CPSs tightly integrates the network field and physical layer, greatly enhancing the interaction between the cyber world and physical elements. Nowadays, CPSs are expanding to every aspect of people's lives, such as unmanned ground vehicle, unmanned aerial vehicles, smart grids (Cao et al., 2025; Chen et al., 2023; Hammad et al., 2019; Lei et al., 2025; Shoukry et al., 2017; Su et al., 2025; Wu et al., 2024a, 2024b; Yang et al., 2024). Due to some unavoidable factors, such as systems subject to some uncertainties or state unmeasurable in some practical applications, the state estimation problem for CPSs has received widespread attention in recent years. For example, the remote state estimation problem was studied for CPSs based on two sensors scheduled with a shared memory channel in Wei and Ye (2023). A secure state estimation problem was addressed based on the designed stochastic modeling framework in Shi et al. (2017), and the estimation results were applied to solve the secure estimation problem of stable linear Gaussian systems. In order to solve the problem that a single sensor cannot fully observe the state of the system, Kalman decomposition techniques and sensor fusion algorithms were presented in Suo et al. (2024). The work in Chang et al. (2018) proposed a secure state estimation approach based on Kalman filter. As is well known, the Kalman filtering method can provide optimal estimates for linear systems with zero mean Gaussian disturbance and noise. However, disturbance and noise may not conform to the Gaussian distribution in practical applications, and their exact statistical information is difficult to obtain. Therefore, how to relax the conservative requirement for disturbance and noise is a valuable topic.

To relax the conservative requirement, an assumption that the disturbance and noise are unknown but bounded (UBB) has been proposed (Ethabet et al., 2018; Liu et al., 2023b; Mazenc & Bernard, 2011; Raïssi et al., 2010; Wang et al., 2022a). Therewith, the interval estimation problem for CPSs with UBB disturbance and noise has been widely investigated in recent years (Fei et al.,

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* Corresponding author.
E-mail addresses: guozhihua0805@163.com (Z. Guo), wangxinjun1991@163.com (X. Wang), tianml163@163.com (M. Tian), hujianing1119@163.com (J. Hu).

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2022; Liu et al., 2023a, 2021; Tang et al., 2020). For example, the interval state estimation problem was investigated for nonlinear systems in Raïssi et al. (2011). In Wang et al. (2019), a set-membership state estimation method was proposed based on the intersection zonotope between the uncertain state set of the system and the measured output set. The work in Ma et al. (2024) designed a fault detection logic based on the obtained interval estimation via reachability analysis. It should be pointed out that the accuracy of interval estimation affects the fault detection rate. However, there are few works on interval estimation considering estimation accuracy. Therefore, it is very necessary to improve the estimation accuracy rate, thereby increasing the fault detection by researching efficient methods.

Overall, DoS attacks are among the most prevalent cyber attacks because they can be executed without obtaining the system's architecture. Such attacks can diminish the dynamic performance of the system and potentially cause instability. Thus, the security problem and stability problem of the system under DoS attacks is one of the hot topics of research in recent years. For example, the sufficient and necessary conditions were proposed for the stability of closed-loop CPSs in the presence of nonperiodic DoS attacks in Lu and Yang (2021). To resist DoS attacks, the work in Zhang and Ye (2020) designed a distributed event-triggered controller based on the proposed new K-connected communication topology construction algorithm. Based on existing results on switched systems (Fei et al., 2023a; Liang et al., 2024; Mancilla-Aguilar & García, 2013; Wu et al., 2015; Zheng et al., 2018), the work in Zhu and Zheng (2020) characterized the periodic DoS attacks by utilizing a cyclic dwell-time switching strategy. Note that from the adversary's point of view, DoS attacks launched by the adversary are random. Nonperiodic DoS attacks more severely disrupt control systems by eliminating regularity, which is more in line with the adversary's intentions. In addition, the above mentioned works do not consider the cumulative attack duration of a DoS attack. It should be pointed out that if the cumulative attack duration of DoS attacks is not considered, it will lead to deviations in system stability and performance analysis, and may even cause catastrophic consequences. This gap motivates us to propose a control method to ensure the stability of the system under long-term or high-frequency nonperiodic DoS attacks.

On the strength of the above discussions, the observer-based switching controller design mechanism and fault detection are investigated for the discrete-time linear CPSs with UBB disturbance and measurement noise under nonperiodic DoS attacks in this paper. The main contributions are arranged as follows:

(1) A new zonotopic observer is designed to estimate the system state for the discrete-time linear CPSs with UBB disturbance and noise in this paper. Unlike the existing works on interval estimation (Fei et al., 2022; Raïssi et al., 2011; Tang et al., 2020; Wang et al., 2019), a decoupled method is proposed to reduce the reachable set of states.

(2) The discrete-time linear CPSs under nonperiodic DoS attacks are transformed into a class of switched systems that includes a stable subsystem in the absence of DoS attacks and an unstable subsystem in the presence of DoS attacks. Based on Lyapunov stability theory, the conditions for the average duration time in the absence of DoS attacks and the average duration time in the presence of DoS attacks are derived to guarantee the globally uniformly exponentially stable (GUES) of the system. Unlike the existing works (Shen et al., 2024; Wang et al., 2022b; Zhao et al., 2022), our work relaxes the assumption conditions related to DoS attacks such that it better aligns with practical engineering applications. In addition, the designed switching law takes into account the cumulative attack duration, which solves the difficulty of system instability under high-frequency or long-term DoS attacks.

(3) An observer-based residual generator is designed to obtain the fault detection logic. Based on the proposed decoupling method and reachability analysis, a tighter residual set is obtained to design more reliable fault detection logic. In addition, through theoretical analysis and simulation verification, the estimation error and the state reachable set obtained are smaller compared to the existing methods in Fei et al. (2022), Raïssi et al. (2011), Tang et al. (2020), Wang et al. (2019).

The rest of this paper is organized below. Section 2 introduces some preliminaries and problem statement. The main results are shown in Section 3. Section 4 discusses the simulation results and Section 5 presents the conclusions.

Notations. The symbols used in this paper are quite standard. $\mathbb{R}^n$ and $\mathbb{R}^{m\times n}$ represent the n dimensional and $m\times n$ dimensional Euclidean space, respectively. $*$ stands for the symmetric block matrix of a symmetry matrix. The superscript T stands for matrix transposition. The symmetric matrix $P \succ 0$ ($P \prec 0$) represents that P is a positive (negative) definite matrix. $I_n \in \mathbb{R}^{n\times n}$ denotes the identity matrix. $\|\bullet\|$ is the Euclidean norm of vector or matrix, and $|\bullet|$ is the absolute value. $\mathscr{L}_2$ represents the space of square summable sequences.

2. Preliminaries and problem statement

An interval $[a, b]$ represents the set $\{x : a \le x \le b\}$, and an unitary interval is defined as $\mathbf{B} = [-1, 1]$. Then, an m-order zonotope $\mathcal{Z} \subset \mathbb{R}^n$ is the affine image of an m-dimensional hypercube $\mathbf{B}^m$ in $\mathbb{R}^n$, i.e.

$$\mathcal{Z} = \langle p, H\rangle = \{p + Hz : z \in \mathbf{B}^m\},$$

where $p \in \mathbb{R}^n$ is the center of $\mathcal{Z}$ and $H \in \mathbb{R}^{n\times m}$ is the generator matrix of $\mathcal{Z}$.

Definition 1 (*Combastel, 2015*)**.** The Minkowski sum of two sets $\mathcal{X}$ and $\mathcal{Y}$ is defined as

$$\mathcal{X} \oplus \mathcal{Y} = \{x + y : x \in \mathcal{X}, y \in \mathcal{Y}\}.$$

Property 1 (*Combastel, 2015*)**.** *For zonotopes, the following equations holds*

$$\begin{aligned} \langle p_1, H_1\rangle \oplus \langle p_2, H_2\rangle &= \langle p_1 + p_2, [H_1 \quad H_2]\rangle, \\ \mathcal{L}\langle p, H\rangle &= \langle \mathcal{L}p, \mathcal{L}H\rangle. \end{aligned} \tag{1}$$

Property 2 (*Tang et al., 2019*)**.** *For a set $\mathcal{X} \subset \mathbb{R}^n$, its interval hull, which is the smallest interval vector that can contain the set $\mathcal{X}$, can be represented as follows:*

$$\mathcal{X} \subseteq \mathbf{Box}(\mathcal{X}) = [\bar{a}, \underline{b}],$$

where $\bar{a} = [a_1, \ldots, a_n]^T$ and $\underline{b} = [b_1, \ldots, b_n]^T$. For a zonotope $\mathcal{Z} \subset \mathbb{R}^n$ with $H \in \mathbb{R}^{n\times s}$, we have $\mathbf{Box}(\mathcal{Z}) = [\bar{a}, \underline{b}]$, where $a_i = p_i - \sum_{j=1}^{s} |H_{i,j}|$ and $b_i = p_i + \sum_{j=1}^{s} |H_{i,j}|$ for $i = 1, \ldots, n$.

Property 3 (*Tang et al., 2019*)**.** *For a series of sets $\mathcal{X}_i$ with $i = 1, \ldots, s$, one has*

$$\mathbf{Box}\Big(\bigoplus_{i=1}^{s} \mathcal{X}_i\Big) = \bigoplus_{i=1}^{s} \mathbf{Box}(\mathcal{X}_i). \tag{2}$$

Lemma 1 (*Tang et al., 2020*)**.** *Given matrices $\mathcal{X} \in \mathbb{R}^{a\times b}$, $\mathcal{Y} \in \mathbb{R}^{b\times c}$ and $\mathcal{Z} \in \mathbb{R}^{a\times c}$, if rank$(\mathcal{Y}) = c$, then the general solution to $\mathcal{X}\mathcal{Y} = \mathcal{Z}$ exists and is given by*

$$\mathcal{X} = \mathcal{Z}\mathcal{Y}^{\dagger} + \mathcal{W}(I_b - \mathcal{Y}\mathcal{Y}^{\dagger}), \tag{3}$$

where $\mathcal{W} \in \mathbb{R}^{a\times b}$ is an arbitrary matrix, $[\bullet]^{\dagger}$ is the Moore–Penrose pseudoinverse.

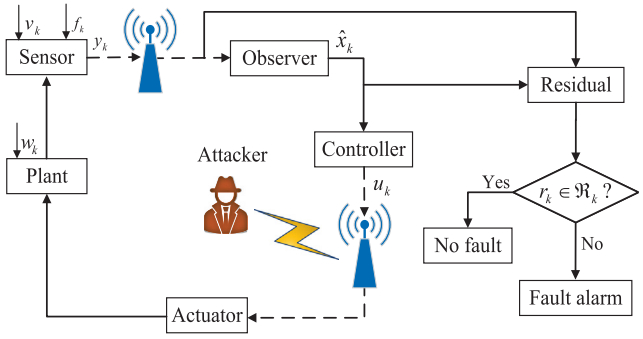


Fig. 1. System structure with fault detection under DoS attacks.

Definition 2 (*Zhang & Shi, 2008*). The equilibrium $x_k = 0$ of the switched system $x_{k+1} = f_{i_k}(x_k, d_k)$ with $d_k = 0$ is GUES under the switching law i_k if there exist constants $\iota > 0$ and $0 < \chi < 1$ such that the solution $x_k = 0$ of the system satisfies $\|x_k\| \le \iota\chi^{(k-k_0)}\|x_{k_0}\|$ for $\forall k \ge k_0$.

Definition 3 (*Xiang et al., 2019*). The switched system

$$\begin{cases} x_{k+1} = \mathcal{F}_{i_k}x_k + \mathcal{G}_{i_k}d_k, \\ y_k = \mathcal{M}_{i_k}x_k + \mathcal{N}_{i_k}d_k, \end{cases} \tag{4}$$

where x_k is the state vector, y_k is the output vector, d_k is the uncertain disturbance, $\mathcal{F}_{i_k}$, $\mathcal{G}_{i_k}$, $\mathcal{M}_{i_k}$ and $\mathcal{N}_{i_k}$ are constant matrices with appropriate dimensions. Given a scalar $\gamma > 0$, the switched system (4) is said to be GUES with an l_2-gain performance, if the switched system (4) is GUES under zero initial condition, and the following inequality holds for all nonzero $d_k \in l_2[0, \infty)$:

$$\sum_{k=0}^{\infty} y_k^T y_k \le \gamma^2 \sum_{k=0}^{\infty} d_k^T d_k. \tag{5}$$

In this paper, we consider the following discrete-time linear CPSs

$$\begin{cases} x_{k+1} = Ax_k + Bu_k + Ew_k, \\ y_k = Cx_k + Df_k + Fv_k, \end{cases} \tag{6}$$

where $x_k \in \mathbb{R}^{n_x}$ is the state vector, $u_k \in \mathbb{R}^{n_u}$ is the input vector, $y_k \in \mathbb{R}^{n_y}$ is the output vector, w_k is the uncertain disturbance, v_k is the measurement noise, $f_k \in \mathbb{R}^{n_f}$ stands for the sensor fault, A, B, C, D, E and F are the system matrices with appropriate dimensions and satisfy $\text{rank}(CE) = n_w$.

In addition, we assume that $f(k) \in \mathscr{L}_2$, x_0, w_k and v_k are UBB as

$$x_0 \in \langle p_0, H_0 \rangle, \ w_k \in \mathbf{W} = \langle 0, H_w \rangle, \ v_k \in \mathbf{V} = \langle 0, H_v \rangle, \tag{7}$$

where $p_0 \in \mathbb{R}^{n_x}$, $H_0 \in \mathbb{R}^{n_x \times n_x}$, $H_w \in \mathbb{R}^{n_w \times n_w}$ and $H_v \in \mathbb{R}^{n_v \times n_v}$ are known vector and matrices.

As shown in Fig. 1, the adversary attempts to deteriorate system performance by disrupting communication networks in practical applications. Due to resource constraints, DoS attacks generally interrupt communication networks in an intermittent manner. Similar to Li et al. (2024), Li and Tong (2023), Pan et al. (2022), Peng and Sun (2020), the intermittent DoS attacks are considered for the system (6) in this paper, and we assume that the attacks occur only between the controller and the actuator. We let $\{k_d\}_{d\in\mathbb{N}}$ represent the time instants of attack initiation and τ_d represents the duration of the dth DoS attack. Therefore, the time interval of the dth DoS attack can be represented by $\mathbb{K}_d := [k_d, k_d + \tau_d)$. Then, let

$$\Xi(k_1, k_2) := \bigcup_{d\in\mathbb{N}} \mathbb{K}_d \bigcap [k_1, k_2]$$

represent the set of time intervals of successfully launched attacks on the interval $[k_1, k_2]$, and let

$$\Lambda(k_1, k_2) := [k_1, k_2] \setminus \Xi(k_1, k_2)$$

represent the set of time intervals of silence for the attacker. To describe the characteristics of intermittent DoS attacks, the following assumption is given.

Assumption 1. Let $N_f(k_2, k_1)$ and $N_n(k_2, k_1)$ denote the number of DoS attacks off/on within the interval $[k_1, k_2]$, respectively. For any $k_2 > k_1 > 0$, there exist constant $\eta_0 \ge 0$, $\tau_{\sigma f} > 0$ and $\tau_{\sigma n} > 0$ such that

$$N_f(k_2, k_1) \le \eta_0 + \frac{\Lambda(k_1, k_2)}{\tau_{\sigma f}(k)}, \tag{8}$$

$$N_n(k_2, k_1) \ge \eta_0 + \frac{\Xi(k_1, k_2)}{\tau_{\sigma n}(k)}, \tag{9}$$

where $\tau_{\sigma f}(k)$ represents the mode-dependent average duration time of attack-free, while $\tau_{\sigma n}(k)$ denotes the mode-dependent average duration time of DoS attacks.

Remark 1. Different from the existing works on periodic DoS attacks (Hu et al., 2019; Tian et al., 2020; Zhu & Zheng, 2020), nonperiodic DoS attacks are studied in this paper. Unlike periodic DoS attacks, the randomness of nonperiodic DoS attacks launched by adversary more severely disrupt control systems by removing predictable patterns, which is more in line with the adversarys intentions.

3. Main results

3.1. State observer design

In this section, a new state observer structure is designed to estimate the state of the system (6). Firstly, the state x_k can be represented as

$$\begin{aligned} x_k &= x_k - QCx_k + QCx_k \\ &= (I_{n_x} - QC)(Ax_{k-1} + Bu_{k-1} + Ew_{k-1}) + QCx_k. \end{aligned} \tag{10}$$

To decouple the disturbance from the estimation error dynamics, Q is chosen to satisfy

$$(I_{n_x} - QC)E = \mathbf{0}_{n_x \times n_w}. \tag{11}$$

The Eq. (11) can be rewritten as

$$QCE = E. \tag{12}$$

Based on Lemma 1, the solution for (12) is

$$Q = E(CE)^{\dagger} + \mathcal{Y}\left(I_{n_y} - (CE)(CE)^{\dagger}\right), \tag{13}$$

where $\mathcal{Y} \in \mathbb{R}^{n_x \times n_y}$ is a designed constant matrix.

We let $i_k \in M = \{0, 1\}$ be the switching law, and its value is represented as

$$i_k = \begin{cases} 1, k \in \Xi(k_1, k_2); \\ 0, k \in \Lambda(k_1, k_2). \end{cases}$$

Then, we define a switching function as follows:

$$\mathbb{U}_{i_k} = \begin{cases} 1, i_k = 1; \\ 0, i_k = 0. \end{cases}$$

Thus, the control signal can be represented as follows:

$$u_k = (1 - \mathbb{U}_{i_k})\bar{u}_k, \tag{14}$$

where $\bar{u}_k$ is a process variable and will be designed subsequently.

From (6) and (10)–(14), the state observer and the controller are designed as follows:

$$\begin{cases} \hat{x}_{k+1} = (I_{n_x} - QC)(A\hat{x}_k + Bu_k) + L(y_k - C\hat{x}_k) \\ \qquad\quad + Qy_{k+1}, \\ u_k = (1 - \mathbb{U}_{i_k})\bar{u}_k, \\ \bar{u}_k = -G\hat{x}_k, \end{cases} \tag{15}$$

where $\hat{x}_k$ is the state estimate of the system, L and G are the observer gain and the controller gain, respectively. The optimized controller gain is presented as

$$G^* = (R + B^T P_c B)^{-1} B^T P_c A, \tag{16}$$

where $P_c = P_c^T \succeq 0$ and $R = R^T \succ 0$ are constant matrices and solved by the Riccati equation as follows:

$$P_c = Q + A^T P_c A - A^T P_c B(R + B^T P_c B)^{-1} B^T P_c A, \tag{17}$$

where $Q = Q^T \succeq 0$ is a given constant matrix.

The estimation error can be derived as

$$e_k = x_k - \hat{x}_k. \tag{18}$$

According to (6) and (15), the estimation error (18) can be obtained that

$$\begin{aligned} e_{k+1} = {} & \big((I_{n_x} - QC)A - LC\big)e_k - LDf_k - LFv_k \\ & - QDf_{k+1} - QFv_{k+1}. \end{aligned} \tag{19}$$

Define the augmented system $\Phi_k \triangleq [x_k^T \quad e_k^T]^T$. Based on (6) and (19), it can be represented as a class of switched systems as follows:

$$\begin{cases} \Phi_{k+1} = \mathcal{A}_{i_k}\Phi_k + \mathcal{B}d_k, \\ y_k = \mathcal{C}\Phi_k + \mathcal{D}d_k, \end{cases} \tag{20}$$

where

$$\mathcal{A}_{i_k} = \begin{bmatrix} A + (1 - \mathbb{U}_{i_k})BG^* & -(1 - \mathbb{U}_{i_k})BG^* \\ 0 & (I_{n_x} - QC)A - LC \end{bmatrix},$$

$$\mathcal{B} = \begin{bmatrix} E & 0 & 0 & 0 & 0 \\ 0 & -LF & -QF & -LD & -QD \end{bmatrix}, \mathcal{C} = \begin{bmatrix} C & 0 \end{bmatrix},$$

$$\mathcal{D} = \begin{bmatrix} 0 & F & 0 & D & 0 \end{bmatrix}, d_k = \begin{bmatrix} w_k^T & v_k^T & v_{k+1}^T & f_k^T & f_{k+1}^T \end{bmatrix}^T.$$

Remark 2. From (20), we can clearly see that it is a class of switched systems. The switching moment of switched systems (20) is the off/on moment of nonperiodic DoS attacks, which delicately describes DoS attacks behavior. Thus, the switching control theory can be employed to solve the observer-based control problem and to ensure the stability of the overall system.

Remark 3. Different from the most of the existing works on interval estimation (Fei et al., 2023b; Tang et al., 2022; Wang et al., 2024), the estimation error dynamics (19) is decoupled from the disturbance based on (11), which can improve estimation accuracy and obtain more accurate reachable sets.

3.2. Stability analysis

In this section, we will derive the conditions that ensure the stability of the system (20) by searching the average duration time in the absence of DoS attacks and the average duration time in the presence of DoS attacks under Assumption 1.

Theorem 1. *Consider switched systems (20) with $d_k = 0$, suppose that there exists $\mathcal{C}^1$ nonnegative functions $V_{i_k} : \mathbb{R}^n \to \mathbb{R}$, $i_k \in M$ and two class $\mathcal{K}_\infty$ functions κ_1 and κ_2 such that*

(1) for $\forall i_k \in M$,

$$\kappa_1(\|\Phi_k\|) \le V_{i_k}(\Phi_k) \le \kappa_2(\|\Phi_k\|); \tag{21}$$

(2) for $i_k = 0$,

$$V_{i_k}(\Phi_{k+1}) \le \varphi_f V_{i_k}(\Phi_k), \tag{22}$$

where $0 < \varphi_f < 1$ is a parameter;

(3) for $i_k = 1$,

$$V_{i_k}(\Phi_{k+1}) \le \varphi_n(k) V_{i_k}(\Phi_k), \tag{23}$$

where $\varphi_n(k) = \varphi_{n_0} + \alpha T_u(0,k)$, $T_u(0,k)$ is the cumulative attack duration, $\varphi_{n_0} > 1$ and $\alpha > 0$ are constants;

(4) for $i_k = 1$ and $i_{k^-} = 0$,

$$V_{i_k}(\Phi_k) \le \eta_f(k) V_{i_{k^-}}(\Phi_{k^-}), \tag{24}$$

where $\eta_f(k) = \eta_{f_0} e^{-\beta T_u(0,k)}$, $\eta_{f_0} > 1$ and $\beta > 0$ are constants;

(5) for $i_k = 0$ and $i_{k^-} = 1$,

$$V_{i_k}(\Phi_k) \le \eta_n V_{i_{k^-}}(\Phi_{k^-}), \tag{25}$$

where $0 < \eta_n < 1$ is a parameter;

(6) the cumulative attack duration meets the following conditions

$$T_u(0,k) \le \epsilon k, \tag{26}$$

where $\epsilon < \frac{\beta}{\beta + \alpha\tau_{\sigma f}^*}$ with $\tau_{\sigma f}^* = -\frac{\ln\eta_{f_0}}{\ln\varphi_f}$. Then, the switched system (20) are GUES for any $\tau_{\sigma f}$ and $\tau_{\sigma n}$ given in Assumption 1 satisfying

$$\begin{cases} \tau_{\sigma f}(k) \ge -\dfrac{\beta T_u(0,k) - \ln\eta_{f_0}}{\ln\varphi_f}, k \in \Lambda(0,k); \\ \tau_{\sigma n}(k) \le -\dfrac{\ln\eta_n}{\ln(\varphi_{n_0} + \alpha T_u(0,k))}, k \in \Xi(0,k). \end{cases} \tag{27}$$

Proof. We take the Lyapunov function for switched systems (20) as follows:

$$V_{i_k}(\Phi_k) = \Phi_k^T P_{i_k}(k)\Phi_k, i_k \in M, \tag{28}$$

where $P_{i_k}(k) \in \mathbb{R}^{2n_x \times 2n_x} \succ 0$. When $d_k = 0$, we can obtain by integrating (21)–(25) as

$$\begin{aligned} V_{i_k}(\Phi_k) &\le \varphi_{i_{k_j}}^{k-k_j} V_{i_{k_j}}(\Phi_{k_j}) \\ &\le \eta_{i_{k_j}} \varphi_{i_{k_j}}^{k-k_j} V_{i_{k_j^-}}(\Phi_{k_j^-}) \\ &\le \eta_{i_{k_j}} \varphi_{i_{k_j}}^{k-k_j} \varphi_{i_{k_{j-1}}}^{k_j-k_{j-1}} V_{i_{k_{j-1}}}(\Phi_{k_{j-1}}) \\ &\cdots \\ &\le \prod_{m=1}^{j} \eta_{i_{k_m}} \varphi_{i_{k_j}}^{k-k_j} \varphi_{i_{k_{j-1}}}^{k_j-k_{j-1}} \cdots \varphi_{i_{k_0}}^{k_1-k_0} V_{i_{k_0}}(\Phi_0), \end{aligned} \tag{29}$$

where $\eta_{i_{k_m}} = \eta_n$ when $i_{k_j} = 0$ and $i_{k_j^-} = 1$; on the contrary, $\eta_{i_{k_j}} = \eta_f$.

Since the system is subject to the intermittent DoS attacks that satisfies Assumption 1, we can get that

$$\begin{aligned} & V_{i_k}(\Phi_k) \\ & \le \eta_{f_0}^{N_{\sigma f}(k,0)} e^{-\beta T_u(0,k) N_{\sigma f}(k,0)} \eta_n^{N_{\sigma n}(k,0)} \varphi_f^{\Lambda(0,k)} \big(\varphi_{n_0} \\ & \quad + \alpha T_u(0,k)\big)^{\Xi(0,k)} V_{i_k}(\Phi_0) \\ & \le \exp\Big\{\Lambda(0,k)\Big(\frac{\ln\eta_{f_0} - \beta T_u(0,k)}{\tau_{\sigma f}(k)} + \ln\varphi_f\Big) + \Xi(0,k)\Big(\frac{\ln\eta_n}{\tau_{\sigma n}(k)} \\ & \quad + \ln(\varphi_{n_0} + \alpha T_u(0,k))\Big) + \eta_0 \ln\eta_{f_0} + \eta_0 \ln\eta_n \\ & \quad - \eta_0 \beta T_u(0,k)\Big\} V_{i_k}(\Phi_0). \end{aligned} \tag{30}$$

Based on Lyapunov stability theory, (27) can be obtained. Through the principle of limits, it can be derived that $\epsilon < \frac{\beta}{\beta+\alpha\tau_{\sigma f}^*}$, $k \to \infty$. Based on Assumption 1 and (27), it can be deduced that $V_{i_k}(\Phi_k)$ converges to 0 as $k \to \infty$. Finally, the GUES can be deduced by combining (21) with Definition 2. The proof is completed. ■

Remark 4. Different from the existing works on DoS attacks (Shen et al., 2024; Wang et al., 2022b; Zhao et al., 2022), the switching law designed in this paper takes into account the cumulative attack duration $T_u(0, k)$. Over time, it can be reasonably assumed from (27) that the mode-dependent average duration time $\tau_{\sigma f}(k)$ of attack-free will increase, the attacker may adjust its attack strategy. For instance, the attacker may reduce the frequency or duration of attacks to avoid being detected or to conserve resources. As a result, the mode-dependent average duration time $\tau_{\sigma f}(k)$ of attack-free will increase and the mode-dependent average duration time $\tau_{\sigma n}(k)$ of DoS attacks will decrease. The increase in the attack-free duration provides the system with more time for stable operation and recovery, while the decrease in the DoS attack duration reduces the interference caused by attacks. Therefore, the stability of the system is guaranteed in long-term or high-frequency nonperiodic DoS attacks.

3.3. l_2-Gain performance analysis

Theorem 2. *Consider switched systems (20), for given scalar $\gamma > 0$, if there exist constants $0 < \varphi_f < 1$ and $0 < \eta_n < 1$, matrices $P_{i_k}(k) = P_{i_k}^T(k) \in \mathbb{R}^{2n_x \times 2n_x} \succ 0$, $i_k \in M$ such that*

$$P_{i_k}(k+1) \le \varphi_f P_{i_k}(k), i_k = 0, \tag{31}$$

$$P_{i_k}(k+1) \le \varphi_n(k) P_{i_k}(k), i_k = 1, \tag{32}$$

$$P_{i_k}(k) \le \eta_f(k) P_{i_{k^-}}(k^-), i_k = 1 \text{ and } i_{k^-} = 0, \tag{33}$$

$$P_{i_k}(k) \le \eta_n P_{i_{k^-}}(k^-), i_k = 0 \text{ and } i_{k^-} = 1, \tag{34}$$

$$\begin{bmatrix} -P_{i_k}(k) & * & * & * \\ 0 & -\gamma^2 I & * & * \\ P_{i_k}(k+1)\mathcal{A}_{i_k} & P_{i_k}(k+1)\mathcal{B} & -P_{i_k}(k+1) & * \\ \mathcal{C} & \mathcal{D} & 0 & -I \end{bmatrix} \prec 0, \tag{35}$$

then for any $\tau_{\sigma f}$ and $\tau_{\sigma n}$ satisfying the conditions (27), switched systems (20) with $d_k = 0$ are GUES and have an l_2-gain γ.

Proof. The GUES can be achieved based on (27) and (31)–(34). In the rest of the proof, we will focus on proving the l_2-gain performance. First, a function is defined as follows:

$$J = \sum_{k=0}^{\infty} [y_k^T y_k - \gamma^2 d_k^T d_k]. \tag{36}$$

Let $\Lambda_{i_k}(k) = y_k^T y_k - \gamma^2 d_k^T d_k + \Delta V_{i_k}(\Phi_k)$. Then, it can be obtained as

$$\begin{aligned} J &= \sum_{m=0}^{\infty} \left(\sum_{k=\delta_m}^{\delta_{m+1}-1} \Lambda_{i_k}(k) - V_{i_k}(\Phi_{\delta_{m+1}}) + V_{i_k}(\Phi_{\delta_m}) \right) \\ &= \sum_{m=0}^{\infty} \left(\sum_{k=\delta_m}^{\delta_{m+1}-1} \Lambda_{i_k}(k) \right) + \sum_{m=1}^{\infty} \left(V_{i_k}(\Phi_{\delta_m}) - V_{j_k}(\Phi_{\delta_m}) \right), \end{aligned} \tag{37}$$

where δ_m is the switching instants, $i_k, j_k \in M$ and $i_k \neq j_k$. Utilizing the Schur complement lemma, (35) is converted to

$$\Omega_{i_k} = \begin{bmatrix} \Omega_{11} & \mathcal{A}_{i_k}^T P_{i_k}(k+1)\mathcal{B} + \mathcal{C}^T\mathcal{D} \\ * & \mathcal{B}^T P_{i_k}(k+1)\mathcal{B} + \mathcal{D}^T\mathcal{D} - \gamma^2 I \end{bmatrix} \prec 0, \tag{38}$$

where $\Omega_{11} = \mathcal{A}_{i_k}^T P_{i_k}(k+1)\mathcal{A}_{i_k} - P_{i_k}(k) + \mathcal{C}^T\mathcal{C}$. It is not hard to see that $\Lambda_{i_k}(k) < 0$ since $\Lambda_{i_k}(k) = \vartheta_k^T \Omega_{i_k} \vartheta_k$ with $\vartheta_k = [\Phi_k^T \quad d_k^T]^T$. Furthermore, we can also obtain that $V_{i_k}(\Phi_{\delta_m}) - V_{j_k}(\Phi_{\delta_m}) < 0$ based on (33) and (34). Thus, $J < 0$ can be ensured. As a result, based on Definition 3, the l_2-gain performance can be achieved, which completes the proof. ■

Remark 5. With the aid of the Lyapunov method under the switching strategy, the stability condition derived in Theorem 2 ensures the desired performance index. In practical applications, we expect the longer the duration time in the absence of DoS attacks and the shorter the duration time in the presence of DoS attacks. Therefore, in order to achieve this goal and guarantee the stability of the overall system, we have derived the average duration time in the absence of DoS attacks and the average duration time in the presence of DoS attacks under Assumption 1.

3.4. H_∞ performance analysis and reachability analysis

Based on H_∞ technique, we construct the observer (15) such that the error dynamics (19) are robust, and then tighten the bound of e_k to obtain more accurate interval estimates. Therefore, the following theorem is presented.

Theorem 3. *Assume that $f_k = 0$ and the conditions of Theorem 1 are satisfied. For given scalars $\gamma > 0$ and $\bar{\gamma} > 0$, if there exist constants $0 < \varphi_f < 1$, $0 < \eta_n < 1$ and $\nu > 1$, matrices $S \in \mathbb{R}^{n_x \times n_y}$, $\mathcal{Y} \in \mathbb{R}^{n_x \times n_y}$, $P \in \mathbb{R}^{n_x \times n_x} \succ 0$ and $P_{i_k}(k) = P_{i_k}^T(k) \in \mathbb{R}^{2n_x \times 2n_x} \succ 0$, $i \in M$ such that*

$$P_{i_k}(k+1) \le \varphi_f P_{i_k}(k), i_k = 0, \tag{39}$$

$$P_{i_k}(k+1) \le \varphi_n(k) P_{i_k}(k), i_k = 1, \tag{40}$$

$$P_{i_k}(k) \le \eta_f(k) P_{i_{k^-}}(k^-), i_k = 1 \text{ and } i_{k^-} = 0, \tag{41}$$

$$P_{i_k}(k) \le \eta_n P_{i_{k^-}}(k^-), i_k = 0 \text{ and } i_{k^-} = 1, \tag{42}$$

$$\begin{bmatrix} -P_{i_k}(k) & * & * & * \\ 0 & -\gamma^2 I & * & * \\ P_{i_k}(k+1)\mathcal{A}_{i_k} & P_{i_k}(k+1)\mathcal{B} & -P_{i_k}(k+1) & * \\ \mathcal{C} & \mathcal{D} & 0 & -I \end{bmatrix} \prec 0, \tag{43}$$

$$\begin{bmatrix} I_{n_x} - P & * & * \\ 0 & -\bar{\gamma}^2 I_{2n_v} & * \\ \Omega_1 & \Omega_2 & -P \end{bmatrix} \prec 0, \tag{44}$$

where $\Omega_1 = PA - PE\Theta^\dagger CA + V\Gamma A - SC$, $\Omega_2 = -SF - PE\Theta^\dagger F - V\Gamma F$, $\Gamma = I_{n_y} - (CE)(CE)^\dagger$, $\Theta = CE$, $L = P^{-1}S$ and $\mathcal{Y} = P^{-1}V$. Then, for any $\tau_{\sigma f}$ and $\tau_{\sigma n}$ satisfying the conditions (27) and Assumption 1, switched systems (20) with $d_k = 0$ are GUES and have an l_2-gain γ, and the observer (15) is a robust observer for the system (6) and the error system (19) satisfies $\|e\| \le \bar{\gamma}\|v\|$.

Proof. The GUES and the l_2-gain performance analysis can be achieved in Theorem 2. In the remaining part of the proof, we mainly focus on obtaining the robust observer by incorporating H_∞ technique.

Based on the Schur complement lemma and (44), one obtains

$$\begin{bmatrix} \tilde{A}^T P \tilde{A} - P + I_{n_x} & * \\ -\mathcal{E}^T P \tilde{A} & \mathcal{E}^T P \mathcal{E} - \bar{\gamma}^2 I_{n_v} \end{bmatrix} \prec 0, \tag{45}$$

where $\tilde{A} = (I_{n_x} - QC)A - LC$ and $\mathcal{E} = [-LF \quad -QF]$. Then, based on the bounded real lemma (Xu & Lam, 2006), the error system (19) is stable and satisfies $\|e\| \le \bar{\gamma}\|v\|$. The proof is completed. ■

The minimal parameter $\bar{\gamma}$ can be attained by solving the following optimization problem:

$$\begin{aligned}&\min \quad \bar{\gamma}^2,\\ &s.t. \quad (43),(44) \text{ and } P \succ 0.\end{aligned}$$

By solving the above optimization problem, the optimal observer gain is attained. Then, via utilizing the reachability analysis of (19), the interval estimate of x_k can be attained according to the following theorem.

Theorem 4. *Assume that $f_k = 0$ and the conditions of Theorem 3 are satisfied. For the studied system (6), the interval estimation of state x_k can be attained based on (15) with $\hat{x}_0 = p_0$ and*

$$\begin{aligned}&\mathbf{Box}(\Psi_k)\\ =&\mathbf{Box}\big(((I_{n_x}-QC)A-LC)^k\langle 0,H_0\rangle\big)\\ &\oplus\bigoplus_{l=0}^{k-1}\mathbf{Box}\big(((I_{n_x}-QC)A-LC)^l(-LF-QF)\mathbf{V}\big),\end{aligned}\tag{46}$$

where $\mathbf{Box}(\Psi_k)=[\underline{e}_k,\bar{e}_k]$ with $\mathbf{Box}(\langle 0,H_0\rangle)=[\underline{e}_0,\bar{e}_0]$.

Proof. From (19), one can get that

$$\begin{aligned}e_k=&\big((I_{n_x}-QC)A-LC\big)^k e_0\\ &+\sum_{l=0}^{k-1}\big((I_{n_x}-QC)A-LC\big)^l\big(-LFv_{k-l-1}\big)\\ &+\sum_{l=0}^{k-1}\big((I_{n_x}-QC)A-LC\big)^l\big(-QFv_{k-l}\big).\end{aligned}\tag{47}$$

Let Ψ_k be the reachable set of e_k, i.e. $e_k\in\Psi_k$. Based on (7) and (47), Ψ_k is derived as follows:

$$\begin{aligned}\Psi_k=&\big((I_{n_x}-QC)A-LC\big)^k\Psi_0\\ &\oplus\bigoplus_{l=0}^{k-1}\big((I_{n_x}-QC)A-LC\big)^l(-LF-QF)\mathbf{V}.\end{aligned}\tag{48}$$

According to $x_0\in\langle p_0,H_0\rangle$ and $\hat{x}_0=p_0$, one can derive that

$$\Psi_0=\langle 0,H_0\rangle.$$

From Property 2, one gets $\Psi_k\in\mathbf{Box}(\Psi_k)=[\underline{e}_k,\bar{e}_k]$. It is not hard to see that $e_k\in\mathbf{Box}(\Psi_k)$ and $\underline{e}_k\le e_k\le\bar{e}_k$. Based on (18), we have $\underline{e}_k+\hat{x}_k=\underline{x}_k\le x_k\le\bar{x}_k=\bar{e}_k+\hat{x}_k$. It completes the proof. ■

Remark 6. Like the existing results on interval estimation (Tang et al., 2020; Yang et al., 2023), the error reachable set is obtained as follows:

$$\begin{aligned}\mathbf{Box}(\Psi_k)=&\mathbf{Box}\big((A-LC)^k\langle 0,H_0\rangle\big)\\ &\oplus\bigoplus_{l=0}^{k-1}\mathbf{Box}\big((A-LC)^lE\mathbf{W}\big)\\ &\oplus\bigoplus_{l=0}^{k-1}\mathbf{Box}\big((A-LC)^l\mathcal{E}(-LF-QF)\big).\end{aligned}$$

Different from the interval estimation methods in Fei et al. (2022), Raïssi et al. (2011), Tang et al. (2020), Wang et al. (2019), Yang et al. (2023), a decoupling method is proposed to decouple the disturbance from the error system in this paper, which also separates the obtained reachable set (46) from the disturbance. The goal of obtaining the smaller state reachable set is achieved.

3.5. Fault detection based on zonotopic state observer

In this part, we try to develop a fault detection strategy to detect the occurrence of the fault by combining the designed observer (15) with the consider CPSs (6). To this end, the observer-based residual signal generator is established in the fault-free situation as follows:

$$\begin{aligned}r_k&=y_k-C\hat{x}_k\\ &=Ce_k+Fv_k.\end{aligned}\tag{49}$$

The reachable set $\mathfrak{R}_k$ of the residual signal under fault-free case is calculated as follows:

$$\begin{aligned}r_k&\in\mathfrak{R}_k\\ &=C\Psi_k+F\mathbf{V}\\ &=\big(C((I_{n_x}-QC)A-LC)\big)^k\Psi_0\\ &\quad\oplus\bigoplus_{l=0}^{k-1}\big(C((I_{n_x}-QC)A-LC)\big)^l(-CLF-CQF+F)\mathbf{V}\\ &=\langle p_k^r,H_k^r\rangle,\end{aligned}\tag{50}$$

where $p_k^r=0$ and

$$\begin{aligned}H_k^r=&\big(C((I_{n_x}-QC)A-LC)\big)^kH_0\\ &+\sum_{l=0}^{k-1}\big(C((I_{n_x}-QC)A-LC)\big)^l(-CLF-CQF+F)H_v.\end{aligned}$$

Then, the upper bound $\bar{r}_k$ and lower bound $\underline{r}_k$ of the residual signal r_k can be obtained as follows:

$$\begin{cases}\bar{r}_k=p_k^r+H_k^r,\\ \underline{r}_k=p_k^r-H_k^r.\end{cases}$$

When the system is subjected to sensor faults, the reachable set of the residual signal can be expressed as follows:

$$\begin{aligned}r_k'&=y_k-C\hat{x}_k\\ &=Ce_k+Df_k+Fv_k\\ &\in\left\langle p_k^{r'},H_k^r\right\rangle,\end{aligned}\tag{51}$$

where $p_k^{r'}=p_k^r+Df_k$.

It is clear that the fault f_k affect the center of the reachable set of the residual signal r_k. Based on the above discussion, it is not difficult to see that $r_k\notin\mathfrak{R}_k$ if there exists the sensor fault, and $r_k\in\mathfrak{R}_k$ if there is no sensor fault. To eliminate the false alarm caused by UBB noises, the detect thresholds can be chosen as follows:

$$\begin{aligned}\bar{r}&=\max_{k\ge 0}\left\{p_k^r+H_k^r\right\};\\ \underline{r}&=\min_{k\ge 0}\left\{p_k^r-H_k^r\right\}.\end{aligned}$$

In light of the above, the detection criterion is designed as follows:

$$\begin{cases}r_k\in[\bar{r},\underline{r}], & \text{fault occurrence;}\\ r_k\notin[\bar{r},\underline{r}], & \text{no fault occurrence.}\end{cases}\tag{52}$$

Remark 7. For the obtained reachable sets Ψ_k in (48), the most of the existing works considered the worst case of disturbance and noise in Garbouj et al. (2021), Ma et al. (2024, 2023), which may reduce the fault detection rate. Different from these works (Garbouj et al., 2021; Ma et al., 2024, 2023), this paper gets tighter set Ψ_k(48) by utilizing the proposed decoupling methods such that the tighter set $\mathfrak{R}_k$(50) is obtained, which can improve fault detection rates to a certain degree.

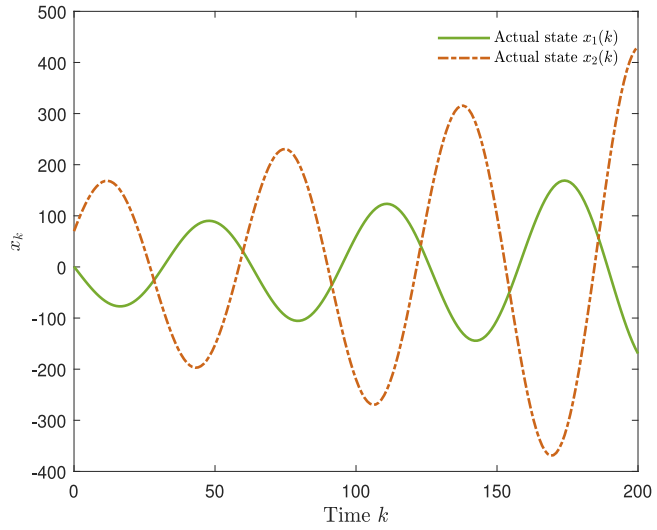


Fig. 2. The system states under $u_i = 0$.

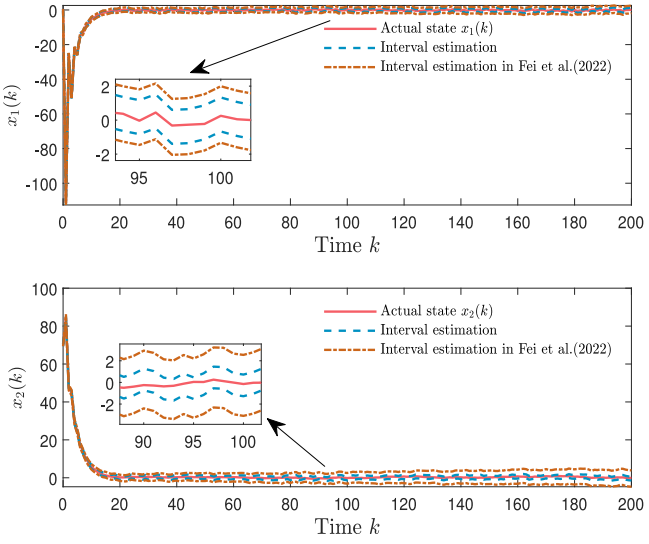


Fig. 3. The actual states and its interval estimation results.

4. Simulation results

In this section, the simulation results are provided to clarify the effectiveness and superiority of the proposed method. The system model (6) has the following parameters:

$$A = \begin{bmatrix} 0.8 & -0.1 \\ 0.5 & 1.2 \end{bmatrix}, B = \begin{bmatrix} 0.25 & 0.95 \\ 0.015 & 0.0007 \end{bmatrix},$$
$$C = \begin{bmatrix} 0.017 \\ 0.04 \end{bmatrix}, D = \begin{bmatrix} 3 & 0 \\ 0 & 8 \end{bmatrix}, E = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$
$$F = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

By solving the Riccati equation and optimization problems, we can obtain as

$$G^* = \begin{bmatrix} 1.3015 & 0.0125 \\ 1.5124 & 1.5542 \end{bmatrix}, L = \begin{bmatrix} 0.0892 & -0.1701 \\ -0.1441 & 0.4368 \end{bmatrix},$$
$$P = \begin{bmatrix} 15.4298 & 5.3999 \\ 5.3999 & 3.0207 \end{bmatrix}, Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

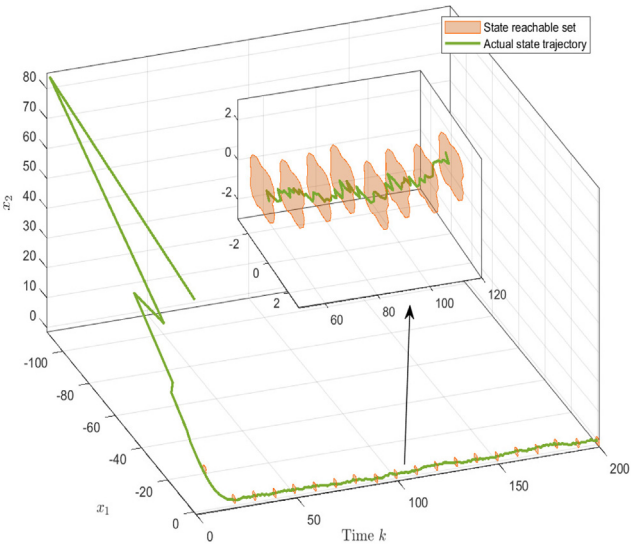


Fig. 4. State reachable set.

The eigenvalues of A are $\lambda_1 = 1.0+0.1i$ and $\lambda_2 = 1.0-0.1i$. A is not Hurwitz stable. It means that if the adversary successfully launches a DoS attack on the communication networks, i.e. $u_k = 0$, the system is unstable. Fig. 2 presents the actual state signals for $u_k = 0$. We can see that the system is divergent if $u_k = 0$. In the simulation, the initial value is given as $x_0^T = [0.57, 70.1]^T$. The parameters are selected as $\eta_{f_0} = 1.8$, $\varphi_f = 0.9$, $\eta_n = 0.2$, $\varphi_{n_0} = 1.2$, $\alpha = 0.01$ and $\beta = 0.1$. In addition, the fault scenario is considered as follows:

$$f_k = \begin{cases} [0, 0]^T, & 0 \le k \ge 70; \\ [0.5, 2.5]^T, & k > 70. \end{cases}$$

Based on the above discussion, the simulation results are plotted in Figs. 3–7. Fig. 3 depicts the interval estimation results under the fault-free scenario of the proposed method in this paper and the existing interval estimation method in Fei et al. (2022), which demonstrates that our proposed method can obtain more accurate interval estimation results. Simulation results of the state reachable set are depicted in Fig. 4. To ensure clear visualization of the evolving reachable sets, we selectively enlarged the zonotope representations at specific time instants. It can be seen from Fig. 4 that the method proposed in this paper can obtain the accurate reachable set of states. Fig. 5 presents the control signal under nonperiodic DoS attacks and fault occurrence cases. Fig. 6 shows the system outputs under nonperiodic DoS attacks and fault occurrence cases. Fig. 7 illustrates the fault detection results of all channels, where 0 and 1 represent that the corresponding channel is fault-free scenario and fault-occurrence scenario, respectively. Fig. 7 shows that all faults can be detected by our proposed fault detection strategy. From the simulation results, it is not hard to see that the system under nonperiodic DoS attacks can achieve stability when the average duration time in the absence of DoS attacks and the average duration time in the presence of DoS attacks satisfy the derived conditions, and the reachable set of the system state can be obtained.

5. Conclusion

In this paper, we have addressed the problems of interval estimation and fault detection for discrete-time linear CPSs subjected to UBB disturbances and noise under nonperiodic DoS attacks and sensor faults. A novel zonotopic observer has been designed to estimate the state of the system with UBB disturbances and noise.

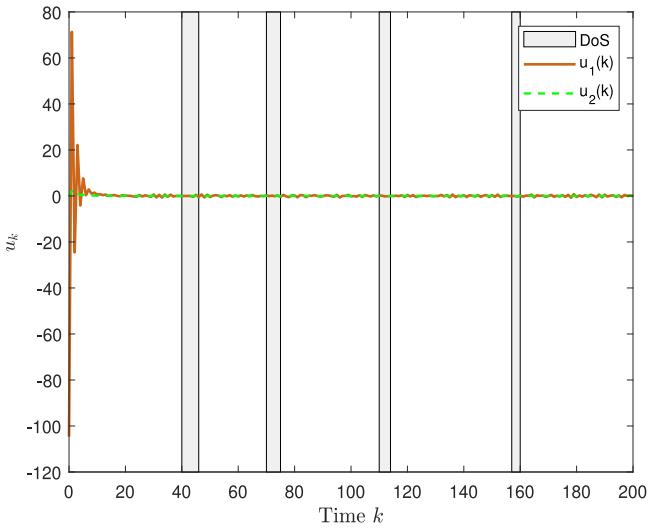


Fig. 5. The control signals u_1 and u_2.

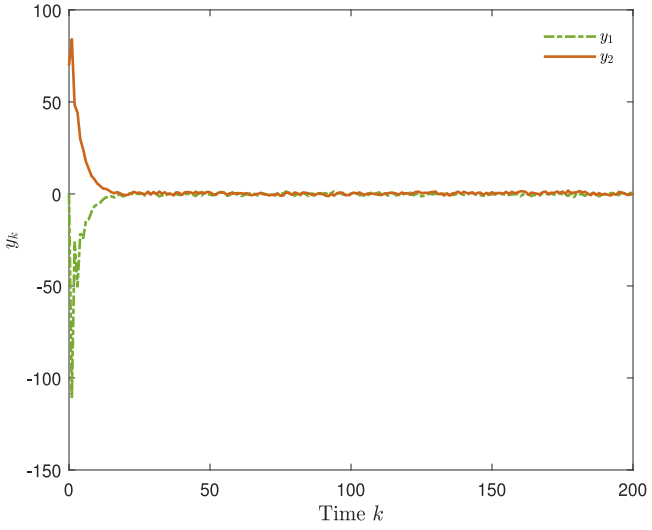


Fig. 6. The system outputs y_1 and y_2.

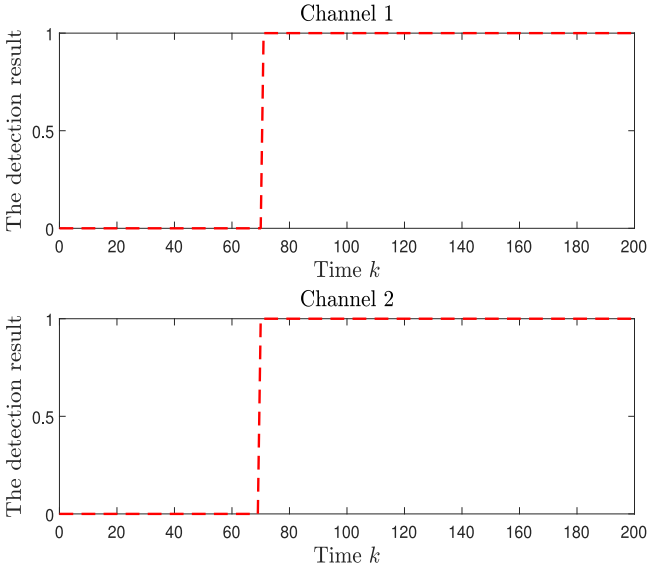


Fig. 7. The fault detection signals.

The CPSs under DoS attacks have been tactfully transformed into a class of augmented discrete-time switched systems, comprising both stable and unstable subsystems. The switching law has been designed to guarantee the stability of the augmented system, which has solved the instability problem of system under long-term or high-frequency DoS attacks. Subsequently, the observer gain has been calculated by solving optimization problems. In addition, the interval state estimation of the system can be obtained through reachability analysis. In addition, we have developed the fault detection mechanism to detect system faults. In future work, we will attempt to design an encryption–decryption mechanism to improve the security of data transmitted over the network.

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Zhihua Guo was born in Henan, China. He received the B.S. degree in automation from Anyang Institute of Technology, Anyang, China, in 2019, and the M.S. degree in control theory and control engineering from Bohai University, Jinzhou, China, in 2022. From March to August 2025, he was a Postgraduate Visiting Internship Student with the Department of Electronic and Computer Engineering at the Hong Kong University of Science and Technology. He is currently pursuing the Ph.D. degree in computer science and technology at Shandong Normal University, Jinan, China. His research interests include cyber–physical systems, set-membership estimation, and security control.



Xinjun Wang was born in Shandong Province, China. He received the M.S. degree in Control Theory and Control Engineering from Bohai University in 2016 and the Ph.D. degree in Information and Communication Engineering from Hohai University in 2020. From September 2022 to July 2023, he worked as a visiting scholar at the School of Automation, Chongqing University. He is currently an Associate Professor with the College of Information Science and Engineering, Shandong Normal University. His research interests include adaptive control theory and application of nonlinear systems, hybrid systems, MASs, and cyber physical systems.



Mingliang Tian was born in Shandong Province, China. He received the B.S. degree in Automation from the School of Automation and Electrical Engineering at Linyi University in 2023. He is currently pursuing his M.S. degree in Control Science and Engineering from the School of Information Science and Engineering at Shandong Normal University. His current research interests include networked control systems, interval state estimation, and fault diagnosis.



Jianing Hu was born in Shandong Province, China. He received the B.Sc. degree in Building Electrical and Intelligence from Zhengzhou University of Light Industry in 2023. He is currently pursuing the M.Sc. degree in Control Science and Engineering at the College of Information Science and Engineering, Shandong Normal University. His research interests include state estimation, cyber–physical system, and privacy-preserving theory.